Effect of DC Bias on Effective Capacitance of MLCCs

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1. Introduction

Opening up any modern computer will reveal hundreds of electrical components, ranging from miniscule resistors to large chips with hundreds of electrical contacts. Among these one shows up on nearly every circuit board. A Multi-Layer Ceramic Capacitor, or MLCC, is used in just about every circuit produced today. Both in precise high speed applications, such as on a CPU, and high power designs, like a DC-DC converter, they can make up half of the components. Through their combination of ease of use in assembly, low cost, and small size, multi-layer ceramic capacitors have become an indispensable tool in circuit design.

Figure 1: MLCCs come in many shapes and sizes, with some as small as a grain of sand

Despite being mainly used in mass production of consumer products, MLCCs are simultaneously used in hobbyist designs. Though MLCCs nearly always come in Surface Mount packages, larger through-hole packages exist, making them easy to solder by hand. While other types of capacitors exist, they do not offer the same characteristics [1].

To see how they function inside of circuits and their purpose, it is necessary to first understand the basic theory of how electricity flows.

1.1. Voltage, Current, and Resistance

Voltage, a potential difference, causes current, the flow of electrons, through a conductor.

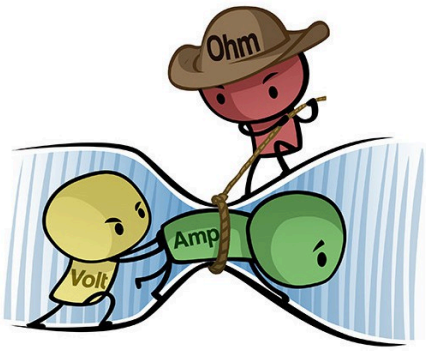


Figure 2: A popular cartoon representing voltage and amperage

When a voltage difference, such as from a battery, is applied across a conductor, current is allowed to flow unrestricted through that conductor. An electrical drawing, known as a schematic, can be used to represent these connections as Figure 3 shows.

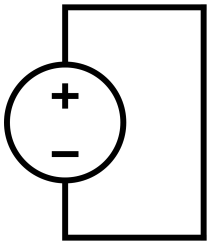


Figure 3: Voltage across conductor

This is known as a short circuit. The amount of current flowing can be limited by using a resistor. A resistor is a component where the current flowing is directly proportional to the amount of voltage across the resistor. This relationship is shown in the following equation where $V =$ voltage (measured in volts), $I =$ current (measured in amperes), and $R =$ resistance (measured in ohms, represented Ω).

$$V = I \cdot R \tag{1}$$

This relationship is known as Ohm's Law [2] and the accompanying circuit can be depicted by a schematic as shown in Figure 4.

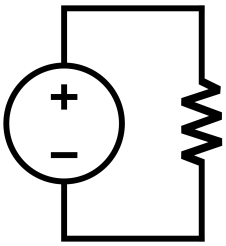


Figure 4: Voltage across resistor

As an example, if a 5V battery is connected to a 100Ω resistor, the current can be found.

$$V = I \cdot R$$

$$5\text{V} = I \cdot 100\Omega$$

$$\frac{5\text{V}}{100\Omega} = I$$

$$0.05\text{A} = I$$

As seen above knowing any two variables of this equation allows solving for the third. The equation can be manipulated to solve for Resistance, $R = \frac{I}{V}$, or current, $I = \frac{V}{R}$.

1.2. Model of Capacitor

A capacitor is a device where the current flowing through is proportional to the rate of change of voltage across the capacitor [3]. That is,

$$I = C\frac{dV}{dt} \tag{2}$$

where C is the capacitance in Farads. When a capacitor is connected to a constant current source the voltage across the capacitor will increase at a constant rate. The schematic of this situation is depicted in Figure 5. Conversely when a capacitor provides a constant current the voltage decreases at a constant rate.

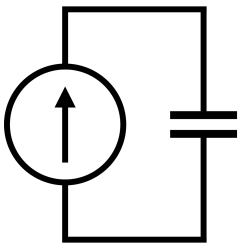


Figure 5: Constant current into capacitor

Functionally, a capacitor acts similarly to a battery in that it can be charged and then discharged.

This models an ideal capacitor, a capacitor that behaves exactly as mathematically modelled, but real capacitors and especially multi-layer ceramic capacitors face non-idealities, which change this model. One of the most important, yet often overlooked, non-idealities in MLCCs is the effect of DC bias. DC bias causes the capacitance, C, to be reduced as a constant voltage is applied across the capacitor [4]. This introduces a new curve, which is the capacitance versus DC bias curve. Manufacturers of capacitors will provide this data [5]. An example of the DC bias graph of a standard capacitor is shown in Figure 6.

These curves have a domain of $0 < V < V_{\text{max}}$ due to maximum ratings and capacitors being bidirectional.

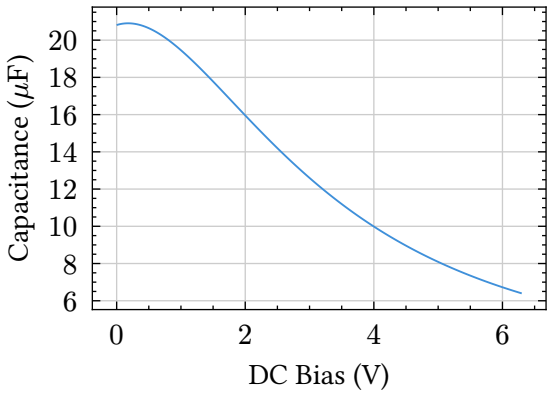


Figure 6: DC Bias of 22μF capacitor from Murata

Because capacitance is in fact a function of voltage, $C = C(V)$, a more accurate model of a capacitor is $I = C(V)\frac{dV}{dt}$. This is important when there is large change in voltage across the capacitor. This effect will be referred to as DC bias throughout this paper.

1.3. Research Question

The effect of DC Bias is mainly used for noise suppression applications. In these situations, because the voltage across the capacitor has a relatively small range, it is sufficient to assume C as constant, functionally making the capacitor a lower-value version of itself. This cannot be done when a device is first turned on, and a MLCC goes from 0V to V_{in}, a large voltage

change. During this transition the capacitor has a large range of capacitance from $C(0V)$ to $C(V_{\text{in}})$.

Due to many systems requiring supply voltage to stabilize before being activated, components often will have a voltage sensing input that will only allow the device to turn on when the voltage reaches a certain threshold. This is often achieved by using a capacitor to limit how fast the voltage on this input rises. Meeting rigid timing requirements necessitates knowing how quickly the voltage across a capacitor will rise. The aim of this paper is to predict the effect DC Bias has on the effective capacitance of a MLCC during large voltage changes such as those found in device turn on.

2. RC Formula

2.1. Applications

Creating the constant current source in Section 1.2 for a capacitor can be difficult, so instead a resistor, as introduced in Section 1.1, can be used to limit voltage increasing across a capacitor. Resistors, similar to capacitors, are relatively cheap and are available in small sizes. Combining these two components creates a circuit often referred to as an RC Circuit. The diagram of this is shown in Figure 7.

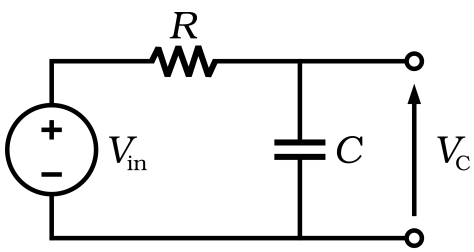


Figure 7: Schematic of RC Circuit

V_C starts at 0V and the current flows through the resistor increasing V_C until it reaches V_{in}.

The voltage across the resistor, V_R, and voltage across the capacitor, V_C, add to V_{in}. Using this, it is possible to mathematically model the circuit as shown below.

$$V_C + V_R = V_{\text{in}}$$

$$V_C = V_{\text{in}} - V_R$$

Using Equation 1 from Section 1.1 for V_R we get:

$$V_C = V_{\text{in}} - IR$$

And because current through the resistor is equivalent to current through the capacitor, we are able to use Equation 2 from Section 1.2 to replace I:

$$V_C = V_{\text{in}} - RC\frac{dV}{dt} \tag{3}$$

The equation in this form is not very helpful because it not possible to directly input a time and get an output voltage. Because of this, the equation is almost always manipulated into the RC formula, shown below. The derivation of Equation 4 is explained in Section 2.2.

$$V_C(t) = V_{\text{in}} \cdot \left(1 - e^{\frac{-t}{RC}}\right) \tag{4}$$

This equation makes it easy to get the output voltage at a specific time in an RC circuit, but only models a circuit where C is constant. Using $R = 10\text{k}\Omega$ and $V_{\text{in}} = 5\text{V}$, a graph of Equation 4 with varying capacitances is shown in Figure 8. These specific values of R and V_{in} were chosen as they will later be used in Section 3.

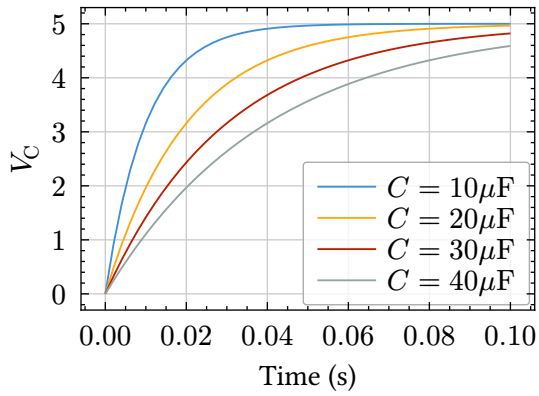


Figure 8: RC Formula with varying capacitances

As seen in Figure 8, a lower capacitance will result in a faster rise time. Because the effect of DC bias causes C to decrease as V_C increases, we are able to predict the voltage will rise faster with a multi-layer ceramic capacitor than its ideal counterpart.

2.2. Derivation

The formula in Section 2.1 can be derived using the following process.

$$V_C = V_{\text{in}} - RC\frac{dV_C}{dt}$$

$$-V_{\text{in}} \qquad -V_{\text{in}}$$

$$V_C - V_{\text{in}} = -RC\frac{dV_C}{dt}$$

$$\div RC \qquad \div RC$$

$$\frac{V_C - V_{\text{in}}}{RC} = -\frac{dV}{dt}$$

$$\cdot\, dt \qquad \cdot\, dt$$

$$\frac{(dt)(V_C - V_{\text{in}})}{RC} = -dV$$

$$\div (V_C - V_{\text{in}}) \qquad \div (V_C - V_{\text{in}})$$

$$\frac{dt}{RC} = \frac{-dV}{V_C - V_{\text{in}}}$$

Take integral

$$\int_0^t \frac{dt}{RC} = \int_0^{V_C} \frac{-dV_C}{V_C - V_{\text{in}}}$$

Integrate

$$\frac{t}{RC} = -\ln(V_C - V_{\text{in}}) - \ln(-V_{\text{in}})$$

Combine logarithms

$$\frac{t}{RC} = -\ln\left(\frac{V_C - V_{\text{in}}}{-V_{\text{in}}}\right)$$

$$\cdot\,(-1) \qquad \cdot\,(-1)$$

$$-\frac{t}{RC} = \ln\left(\frac{V_C - V_{\text{in}}}{-V_{\text{in}}}\right)$$

Logarithm rules

$$e^{-\frac{t}{RC}} = \frac{V_C - V_{\text{in}}}{-V_{\text{in}}}$$

$$\cdot\,(-V_{\text{in}}) \qquad \cdot\,(-V_{\text{in}})$$

$$-V_{\text{in}} \cdot e^{-\frac{t}{RC}} = V_C - V_{\text{in}}$$

$$+(V_{\text{in}}) \qquad +(V_{\text{in}})$$

$$V_{\text{in}} - V_{\text{in}} \cdot e^{-\frac{t}{RC}} = V_C$$

Common factor of V_{in}

$$V_{\text{in}}\left(1 - e^{-\frac{t}{RC}}\right) = V_C$$

3. Experiment

3.1. Overview

As a method of finding electrical behavior, an experiment is the costliest and most time consuming. If the appropriate component needs to be chosen from a large selection, than buying and testing them all is unrealistic.

Despite this shortcoming, an experiment is invaluable in that it will always provide the highest degree of accuracy to how a component will behave in the real world. Due to this, the results from this experiment will be the benchmark for the methods used in Section 4 and Section 5.

3.2. Setup

A printed circuit board designed for a RC circuit, shown in Appendix A, was populated with a 1kΩ resistor and a 22μF capacitor with a rated voltage of 6.3V. The circuit board was then held in place with exposed positive and negative wires. A power supply was set to 5V with no current limit and the negative side of the power supply and PCB were connected. An oscilloscope probe with a ground spring was connected across the capacitor. Another probe was connected between the two power wires. The capacitor was discharged by shorting the two ends using metal tweezers. The positive side of the power supply was then put in contact with the resistor side of the PCB and a waveform was obtained. This setup is depicted in Figure 9. More details on the PCB and oscilloscope used can be found in Appendix A.

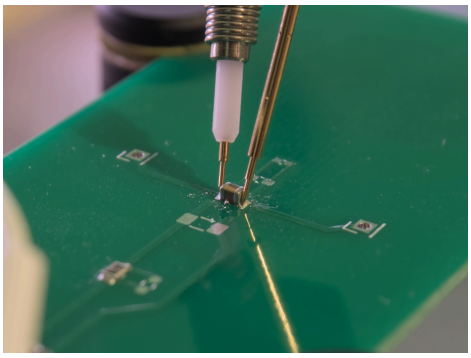


Figure 9: Probing of MLCC using ground spring

3.3. Results

As predicted in Section 2, the voltage rises faster than an ideal 22μF capacitor. A comparison is shown in Figure 10 between an ideal 22μFcapacitor and the real capacitor.

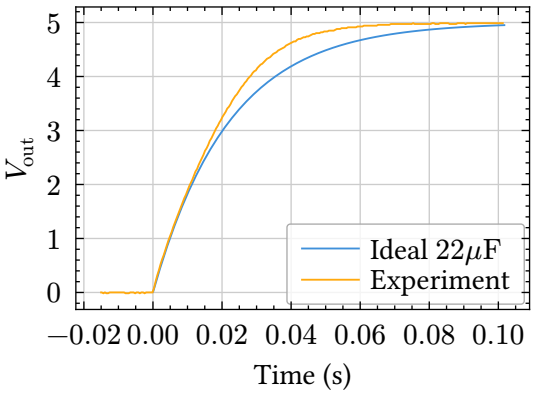


Figure 10: Ideal capacitor compared to experimental results

Using this graph we are able to to try to find a capacitor of a lower value that is more consistent with the experimental results. Using a point on the curve, a capacitor that takes an equivalent amount of time to get to that point can be calculated using the Equation 4. A point of (0.03669s, 4.511V) was chosen and the values of $t = 0.03669\text{s}$ and $V = 4.511\text{V}$ were substituted into Equation 4 as shown below.

$$4.511\text{V} = 5\text{V} \cdot \left(1 - e^{\frac{-0.03669\text{s}}{1000\Omega \cdot C}}\right)$$

Solving for C yielded a result of $C \approx 16\mu$F. The curve from this value of capacitor is plotted in Figure 11.

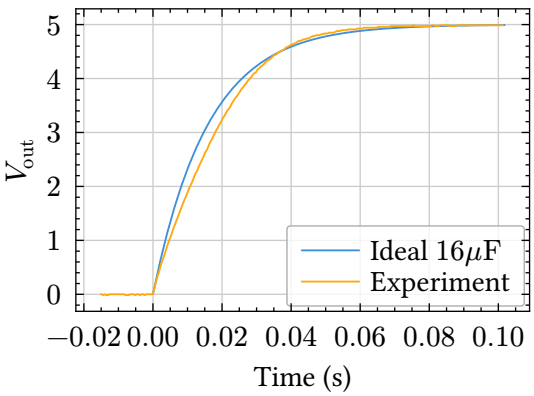


Figure 11: Adjusted capacitance value compared to experiment

The gap between the two curves shows that the original RC formula, even with an adjusted capacitance, is not sufficient for modelling the effects of DC bias.

4. Simulation

4.1. Overview

The speed of modern computing has made simulation in a variety of situations useful. Due to the efficiency of processing repeated operations, computer software is perfect for circuit simulation. In this section, Python, a modern programming language, will be used to simulate an ideal capacitor and then a capacitor with DC bias.

4.2. Implementation

The charging of the capacitor can be broken into small time steps where each time step has value of t_{step}. This will act as a replacement to dt. Each t_{step}, the voltage across the capacitor, V, will increase by a calculated amount. Using this, a sequence of voltage values can be obtained, $V_0, V_1, V_2, ..., V_{\#\text{ of steps}}$. Each V_n corresponds to a time of $t_{\text{step}} \cdot n$.

To find the change in voltage each time step, the equation $I = C\frac{dV}{dt}$ is manipulated to isolate the change in voltage. This yields $dV = I \cdot \frac{dt}{C}$. Current is calculated using Equation 1 to get $I = \frac{V_{\text{in}} - V_n}{R}$. Using these equations we get the following sequence.

$$V_0 = 0V$$

$$V_{n+1} = V_n + \frac{V_{\text{in}} - V_n}{R} \cdot \frac{t_{\text{step}}}{C} \tag{5}$$

All simulations will have a time domain of $0 \leq t < 100\text{ms}$, because the voltage across the capacitor stabilizes by 100ms, as seen in Figure 10.

Using the desired total number of time steps, t_{step} can be calculated with $t_{\text{step}} = \frac{100\text{ms}}{\#\text{ of steps}}$. This is implemented in Python with $R = 1\text{k}\Omega$, $C = 22\mu\text{F}$, and $V_{\text{in}} = 5V$. Code can be found in Appendix A. Figure 12 shows the number of steps as 5, 10, 20, and 50.

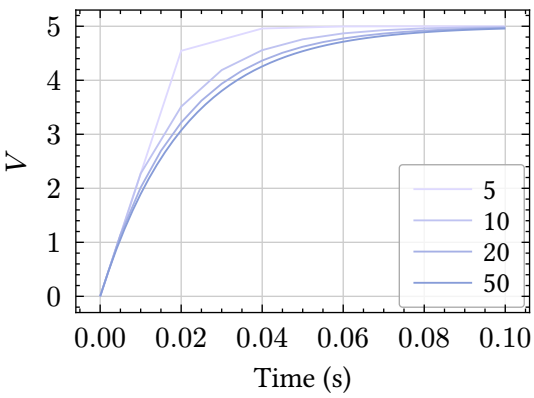


Figure 12: Simulations with increasing number of time steps and decreasing t_{step}

By using this method, the capacitance can easily be changed to be any arbitrary function with time or voltage as the argument. In this case, the capacitance becomes a function of voltage. The new sequence is shown below.

$$V_0 = 0V$$

$$V_{n+1} = V_n + \frac{V_{\text{in}} - V_n}{R} \cdot \frac{t_{\text{step}}}{C(V_n)}$$

Using the DC Bias curve from the manufacturer, the same from Section 1.2, we can use a function to approximate the capacitance based on voltage. The function used was created to linearly decrease from the rated capacitance at $V = 0$ to the end of the DC bias curve. That is, the function intersects (0V, 22μF) and (6.3V, 6.4μF). This function is shown below:

$$C(V) = 22\mu\text{F} - (2.29\mu\text{F} \cdot V) \tag{6}$$

This linear approximation is shown in Figure 13 compared to the curve provided by the manufacturer.

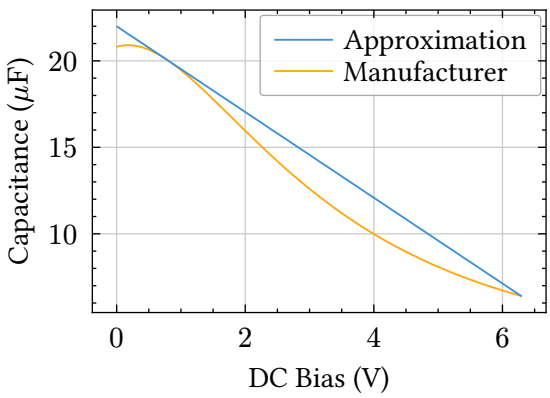


Figure 13: Linear approximation of DC bias

Using this approximation, the sequence becomes

$$V_0 = 0V$$

$$V_{n+1} = V_n + \frac{V_{\text{in}} - V_n}{R} \cdot \frac{t_{\text{step}}}{22\mu\text{F} - 2.29\mu\text{F} \cdot V_n}$$

4.3. Results

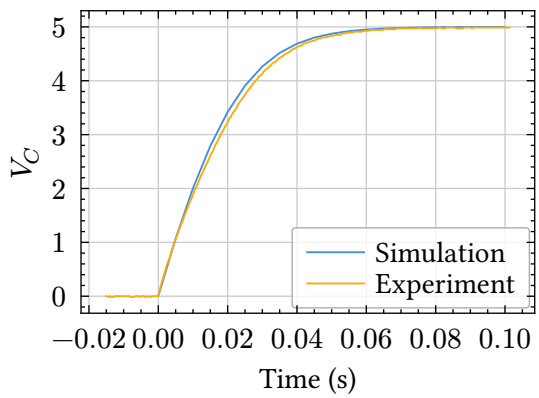


Figure 14: Simulation versus actual

Using this approximation, a simulation is run using 200 time steps across a 100ms period, that is $t_{\text{step}} = 500\mu s$. In Figure 14, curve obtained is compared with the curve obtained from an experiment in Section 3.

This appears to be much closer to the actual capacitor than what would be predicted for an ideal 22μF capacitor.

5. Deriving New Formula

5.1. Overview

In this section we will attempt to derive a formula for the same method used in the second part of the previous section.

In Section 4.2, the approximation $C(V) = 22\mu\text{F} - (2.29\mu\text{F} \cdot V)$ was used. We will continue to use a linear approximation but transfer to an equation form with arbitrary values. $22\mu\text{F}$ and $2.29\mu\text{F}$ are replaced with C_1 and C_2 respectively. Thus the new formula is:

$$C(V) = C_1 - C_2 \cdot V \tag{7}$$

This formula replaces the constant C in Equation 3.

$$V_C = V_{\text{in}} - R(C_1 - C_2 \cdot V_C)\frac{dV_C}{dt}$$

5.2. Results

Using the derivation shown in Appendix B the following equation is found.

$$V_{\text{in}}\left(1 - e^{\frac{C_2 V_C - \frac{t}{R}}{C_1 - C_2 V_{\text{in}}}}\right) = V_C$$

Due to the term $C_2 V_C$ being inside of the exponent, V_C is difficult to isolate, a process out of the scope of this paper. Despite this, time can be found based on voltage. The formula where time is isolated is shown below.

$$t = RC_2 V_C - (R)(C_1 - C_2 V_{\text{in}})\left(\ln\left(\frac{V_{\text{in}} - V_C}{V_{\text{in}}}\right)\right)$$

By calculating t Using a large number of values of V_C in our range of $0V \leq V_C < V_{\text{in}}$ we are able to to obtain a curve. Using $C_1 = 22\mu\text{F}$ and $C_2 = 2.29\mu\text{F}$, a curve is obtained using this method and plotted in Figure 15

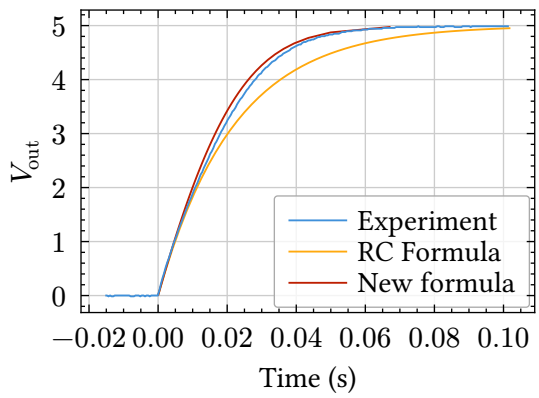


Figure 15: New formula compared to RC formula

As expected, the results of using this formula are functionally identical to the simulation and when compared to the original RC formula, the results are far more accurate

6. Generalization

6.1. Factors Considered

Up to this point, only one capacitor has been analyzed, but the aim of this paper is to be able to accurately predict the behavior of different MLCCs. To do this, the values of C_1 and C_2 in Equation 7 must be able to be generated based on characteristics of a capacitor.

Multi-layer ceramic capacitors have many characteristics, but the most important are:

- Capacitance
- Physical size (eg. 1206, 0805, 0603)
- Maximum voltage
- Temperature coefficient

In this paper only 0805 X5R capacitors will be examined due to being one of the most used package sizes and temperature coefficients.

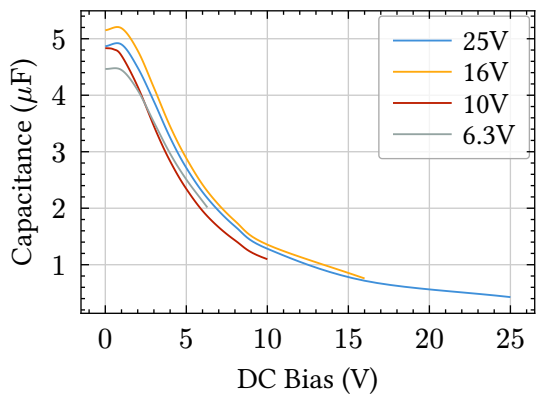


Figure 16: DC bias of different V_{max}

This leaves capacitance and maximum voltage under consideration as factors playing a role in the DC bias characteristics of a capacitor. The second of these, maximum voltage, has a relatively small effect on DC bias as shown in Figure 16. The capacitance, on the other hand, will have a large impact effect on DC bias [6]. Small value MLCCs will have a relatively small change in capacitance across their full voltage range. A 0.5μF capacitor may only change by few percent of its rated capacitance, while a 50 μF capacitor may lose 80%. This effect is shown in Figure 17.

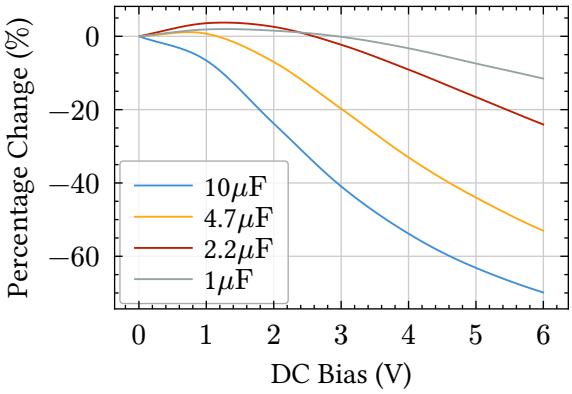


Figure 17: DC bias of varying capacitances

6.2. Approximation

Using two points, like in Section 4, a linear approximation can be constructed for each of the DC bias curves. The first point is (0V, Rated Capacitance) and the second is (6V, C(6V)). The line formed between these two points is shown in Figure 18.

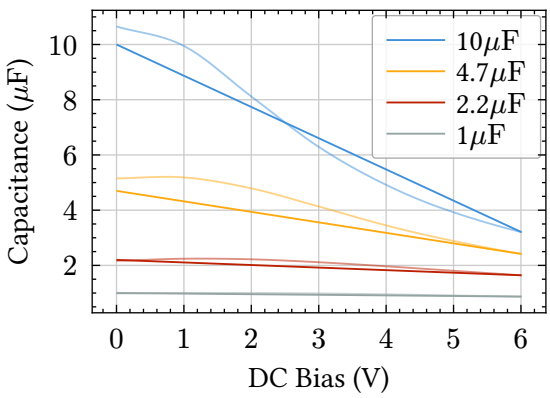


Figure 18: Linear approximation for DC bias of varying capacitances

The slope of each approximation can be found using $\frac{\Delta C}{\Delta V}$. Using the two points this becomes:

$$\text{Slope} = \frac{C(6V) - \text{Rated Capacitance}}{6V - 0V}$$

The slopes of each approximation are plotted in Figure 19. From this, a function to predict the slope of the linear approximation for DC bias can be created. This function was chosen as $f(C) = 0.19 - 0.155C$.

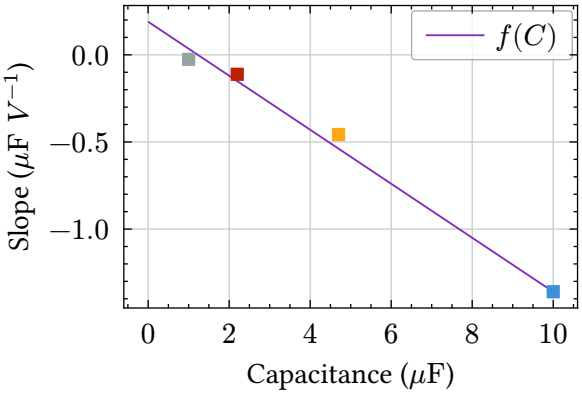


Figure 19: Slope of DC bias curve

At $f(1.23)$ this relationship breaks apart as the slope becomes positive, that is capacitance increases with voltage across a capacitor. In reality, the slope will level off to practically zero as the capacitance approaches zero. A better approximation can be created by using a power function. $g(C) = -0.0352C^{\frac{8}{5}}$ which solves the problem of capacitances close to zero.

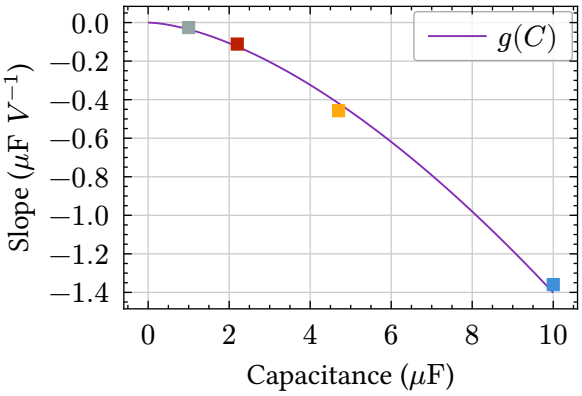


Figure 20: Slope of DC bias curve

Using the rated capacitance as the y-intercept, C_1, and the above equation as the slope, C_2, we find that in general a 0805 X5R MLCC's DC bias

can be approximated with the linear formula shown in Equation 8.

$$C - 0.0352C^{\frac{8}{5}} \tag{8}$$

Combined with the equations derived in Section 5 it is possible to more accurately predict the time it will take for a MLCC used in an RC circuit to rise to a given voltage level.

7. Comparison of Methods

Using Equation 7 for capacitance created the same results for both the simulation and formula, but the simulation offered a greater flexibility. For example, using a simulation would make it possible to model two capacitors with different DC bias curves used in parallel.

While the rated capacitance can be used to estimate the effect of DC bias as shown in Section 6, it is more accurate and practical to refer to the curve from the manufacturer and construct a function for capacitance using capacitor specific data.

The experiment done in Section 3 shows the importance of not relying on the rated capacitance as the 22μF capacitor tested was closer to a 16μF capacitor, a reduction in capacitance of $\approx 27\%$. Despite the importance of DC bias in multi-layer ceramic capacitors, there are other effects that must be taken into consideration such as temperature and aging [4].

8. Conclusion

The creation of this IA did include a few unexpected turns. Originally SPICE (Simulation Program with Integrated Circuit Emphasis) was going to be used, but was replaced by the simulation written in Python due to inflexibilities in the software. The results of this turned out to be quite useful.

In Section 5 I did learn about the *Lambert W Function*, but its complexity limited its use.

All graphs were created using Lilaq, which made data visualization efficient and flexible. Addtional Typst packages used were Codelst, Equate, and Showybox.

A Experimental Setup and Code

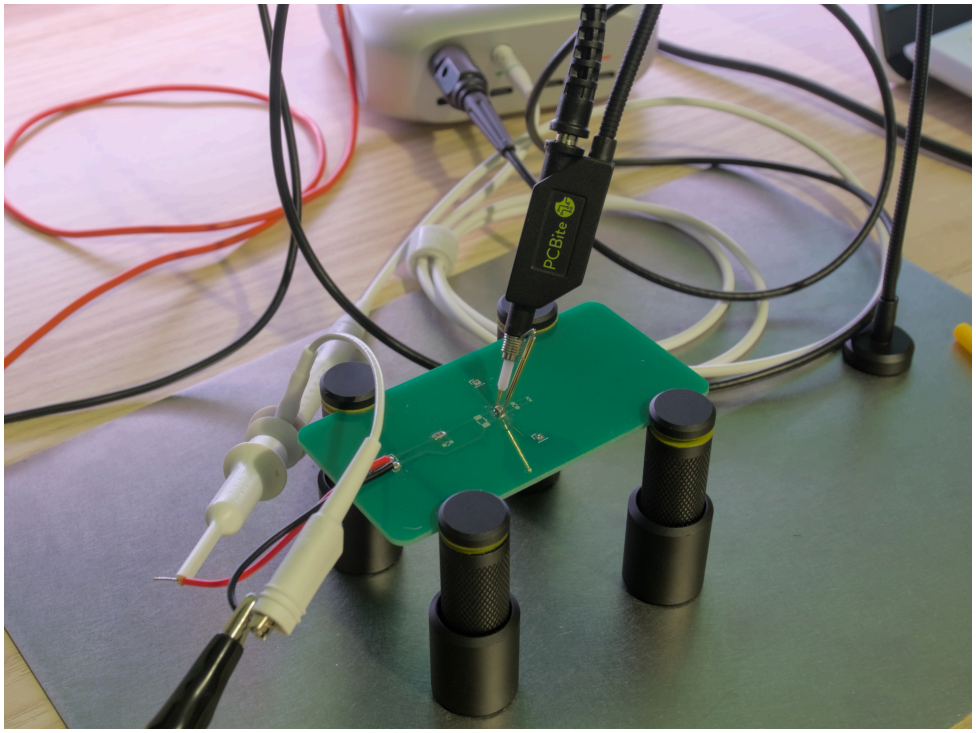


Figure 1: Probing setup

A Saleae Logic MSO oscilloscope was used in combination with a PCBite probe and Saleae probe. Alligator clips from a power supply were used.

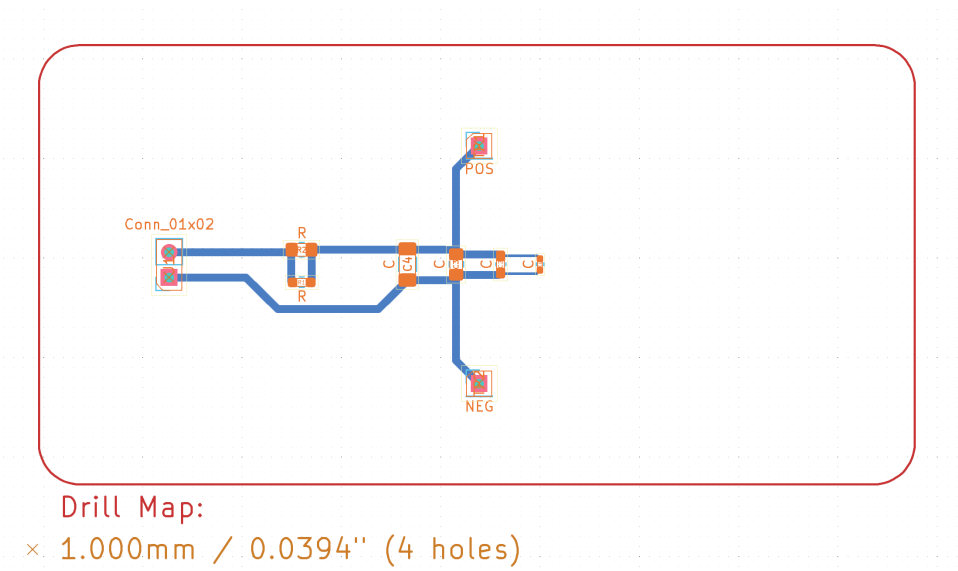
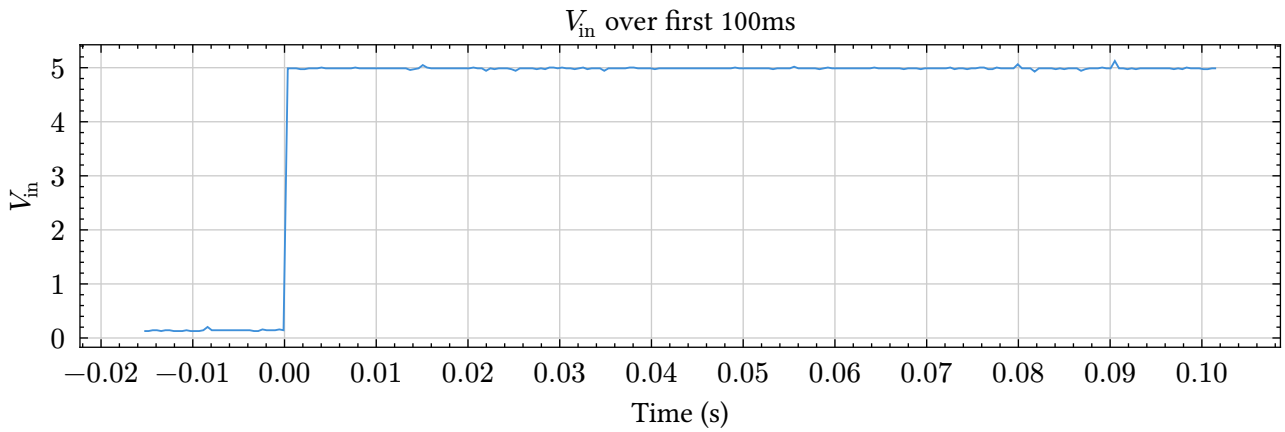
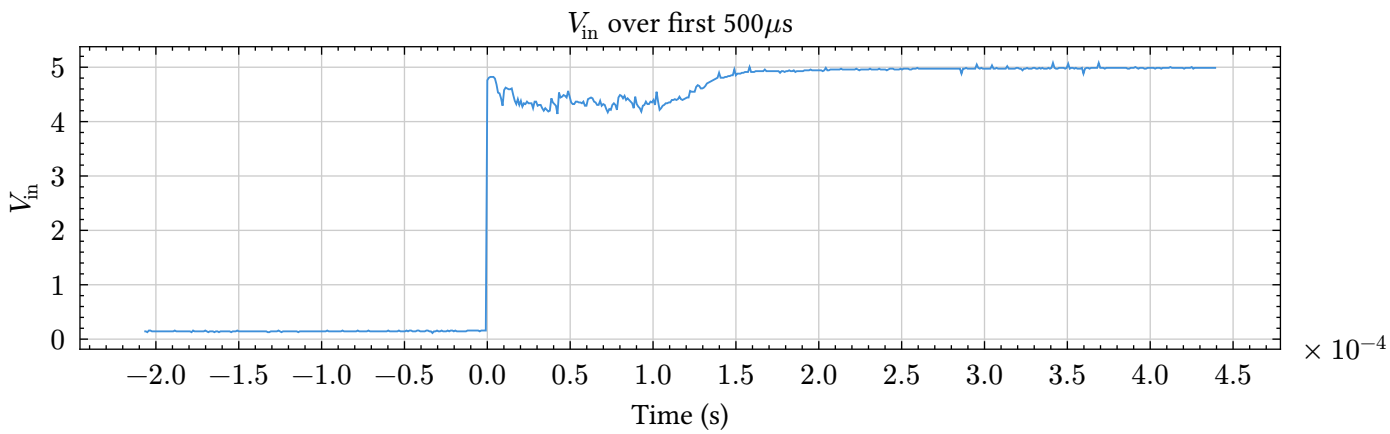


Figure 2: PCB layout


V_{in} over first 100ms
V_{in}
0
1
2
3
4
5
−0.02
−0.01
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.10
Time (s)


V_{in} over first 500μs
V_{in}
0
1
2
3
4
5
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
$\times 10^{-4}$
Time (s)

```python
1  import csv
2
3  file = open("data/sim.csv", mode= "w", newline = "")
4  writer = csv.writer(file)
5
6  #1k ohm
7  R = 1e3
8  #5V
9  V = 5
10 #22uf
11 C = 22e-6
12
13 #100ms
14 end = 0.1
15 steps = 200
16 dt = end/steps
17 t = 0
18
19 #voltage across capacitor
20 Vc = 0
21
22 writer.writerow([0, 0])
23
24 #DC Bias capacitor
25 for i in range(steps):
26     #Voltage and thus current across resistor during this step
27     Vr = V-Vc
28     I = Vr/R
29
30     #Linear approximation of capacitance due to DC bias
31     C_bias = 22e-6 - (2.29e-6 * Vc)
32
33     #Based on current and amount of time, increment voltage
34     dV = (I*dt)/C_bias
35     Vc = Vc + dV
36
37     #record result
38     t = t + dt
39     writer.writerow([t, Vc])
40
41 file.close()
```

This Python script was used to generate data for all five simulation results shown in Section 4 Figure 12 and Figure 14. The variable C in line 34 was changed to C_bias or C to change whether or not to account for DC bias.

```python
import scipy.special as special
import math
import csv

def formula(R, V, C1, C2, t):
    A = C2/(C1-C2*V)
    power = math.exp((-t)/((R*C1) - (R*C2*V)))
    fraction = (A*V) * (math.exp(A*V))
    lambertInner = power * fraction
    lambert = special.lambertw(lambertInner, tol = 1e-9).real
    return V - (lambert * (1/A))

file = open("data/lambert.csv", mode= "w", newline = "")
writer = csv.writer(file)

#1k ohm
R = 1e3
#5V
V = 5

C1 = 22e-6
C2 = 2.29e-6

#100ms
end = 0.1
steps = 20
dt = end/steps
t = 0

writer.writerow([0, 0])

#DC Bias capacitor
for i in range(steps):
    t = t + dt
    result = formula(R, V, C1, C2, t)
    writer.writerow([t, result])

file.close()
```

B Derivations

Deriving Modified RC Formula

$$V_C = V_{\text{in}} - R(C_1 - C_2 \cdot V_C)\frac{dV_C}{dt}$$

$$-(V_{\text{in}})$$

$$V_C - V_{\text{in}} = -R(C_1 - C_2 \cdot V_C)\frac{dV_C}{dt}$$

$$-(V_{\text{in}})$$

$$-dt(V_C - V_{\text{in}}) = R(C_1 - C_2 \cdot V_C)(dV_C)$$

$$-(V_{\text{in}})$$

$$\frac{-dt(V_C - V_{\text{in}})}{R} = (C_1 - C_2 \cdot V_C)(dV_C)$$

$$-(V_{\text{in}})$$

$$\frac{-dt}{R} = \frac{(C_1 - C_2 V_C) \cdot (dV_C)}{V_C - V_{\text{in}}}$$

$$-(V_{\text{in}})$$

$$\int_0^t \frac{-dt}{R} = \int_0^{V_C} \frac{(C_1 - C_2 V_C) \cdot (dV_C)}{V_C - V_{\text{in}}}$$

$$\int_0^t \left(\frac{-dt}{R}\right)$$

$$-(V_{\text{in}})$$

$$-\frac{\boldsymbol{t}}{\boldsymbol{R}}$$

$$\int_0^{V_C} \frac{C_1 - C_2 V_C}{V_C - V_{\text{in}}} dV_C$$

$$-(V_{\text{in}})$$

$$\int_0^{V_C} \frac{C_1}{V_C - V_{\text{in}}} + \frac{-C_2 V_C}{V_C - V_{\text{in}}} dV_C$$

$$-(V_{\text{in}})$$

$$\int_0^{V_C} \frac{C_1}{V_C - V_{\text{in}}} dV_C + \int_0^{V_C} \frac{-C_2 V_C}{V_C - V_{\text{in}}} dV_C$$

$$\int_0^{V_C} \frac{C_1}{V_C - V_{\text{in}}} dV_C$$

$$-(V_{\text{in}})$$

$$C_1 \ln(|V_C - V_{\text{in}}|) - C_1 \ln(|-V_{\text{in}}|)$$

$$-(V_{\text{in}})$$

$$\boldsymbol{C_1 \ln(V_{\text{in}} - V_C) - C_1 \ln(V_{\text{in}})}$$

$$\int_0^{V_C} \frac{-C_2 V_C}{V_C - V_{\text{in}}} dV_C$$

$$-(V_{\text{in}})$$

$$-C_2 \int_0^{V_C} \frac{V_C}{V_C - V_{\text{in}}} dV_C$$

$$-(V_{\text{in}})$$

$$-C_2 \int_0^{V_C} \frac{V_C - V_{\text{in}}}{V_C - V_{\text{in}}} + \frac{V_{\text{in}}}{V_C - V_{\text{in}}} dV_C$$

$$-(V_{\text{in}})$$

$$-C_2 \int_0^{V_C} 1 + \frac{V_{\text{in}}}{V_C - V_{\text{in}}} dV_C$$

$$-(V_{\text{in}})$$

$$-C_2 \int_0^{V_C} 1 dV_C + -C_2 \int_0^{V_C} \frac{V_{\text{in}}}{V_C - V_{\text{in}}} dV_C$$

$$-C_2 \int_0^{V_C} 1 dV_C$$

$$-(V_{\text{in}})$$

$$\boldsymbol{-C_2 V_C}$$

$$-C_2 \int_0^{V_C} \frac{V_{\text{in}}}{V_C - V_{\text{in}}} dV_C$$

$$-(V_{\text{in}})$$

$$-C_2 V_{\text{in}} \int_0^{V_C} \frac{1}{V_C - V_{\text{in}}} dV_C$$

$$-(V_{\text{in}})$$

$$-C_2 V_{\text{in}} \ln(|V_C - V_{\text{in}}|) + C_2 V_{\text{in}} \ln(|-V_{\text{in}}|)$$

$$-(V_{\text{in}})$$

$$\boldsymbol{-C_2 V_{\text{in}} \ln(V_{\text{in}} - V_C) + C_2 V_{\text{in}} \ln(V_{\text{in}})}$$

Combining results

$$\frac{-t}{R} = C_1 \ln(V_{\text{in}} - V_C) - C_1 \ln(V_{\text{in}}) - C_2 V_{\text{in}} \ln(V_{\text{in}} - V_C) + C_2 V_{\text{in}} \ln(V_{\text{in}}) - C_2 V_C$$

$$-(V_{\text{in}})$$

$$\frac{-t}{R} = \ln(V_{\text{in}} - V_C)(C_1 - C_2 V_{\text{in}}) - \ln(V_{\text{in}})(-C_1 + C_2 V_{\text{in}}) - C_2 V_C$$

$$-(V_{\text{in}})$$

$$\frac{-t}{R} = \ln(V_{\text{in}} - V_C)(C_1 - C_2 V_{\text{in}}) + \ln(V_{\text{in}})(C_1 - C_2 V_{\text{in}}) - C_2 V_C$$

$$-(V_{\text{in}})$$

$$\frac{-t}{R} = (C_1 - C_2 V_{\text{in}})(\ln(V_{\text{in}} - V_C) + \ln(V_{\text{in}})) - C_2 V_C$$

$$-(V_{\text{in}})$$

$$\boldsymbol{\frac{-t}{R} = (C_1 - C_2 V_{\text{in}}) \left(\ln \left(\frac{V_{\text{in}} - V_C}{V_{\text{in}}} \right) \right) - C_2 V_C}$$

Manipulating into Lambert W form

$$\frac{-t}{R} = (C_1 - C_2 V_{\text{in}})\left(\ln\left(\frac{V_{\text{in}} - V_C}{V_{\text{in}}}\right)\right) - C_2 V_C$$

$$-(V_{\text{in}})$$

$$\frac{-t}{(R)\cdot(C_1 - C_2 V_{\text{in}})} = \ln\left(\frac{V_{\text{in}} - V_C}{V_{\text{in}}}\right) - \frac{C_2 V_C}{C_1 - C_2 V_{\text{in}}}$$

$$-(V_{\text{in}})$$

$$A = \frac{C_2}{C_1 - C_2 V_{\text{in}}}$$

$$-(V_{\text{in}})$$

$$e^{\frac{-t}{(R)\cdot(C_1 - C_2 V_{\text{in}})}} = e^{\ln\left(\frac{V_{\text{in}} - V_C}{V_{\text{in}}}\right) - AV_C}$$

$$-(V_{\text{in}})$$

$$e^{\frac{-t}{(R)\cdot(C_1 - C_2 V_{\text{in}})}} = e^{\ln\left(\frac{V_{\text{in}} - V_C}{V_{\text{in}}}\right)} e^{-AV_C}$$

$$-(V_{\text{in}})$$

$$e^{\frac{-t}{(R)\cdot(C_1 - C_2 V_{\text{in}})}} = \left(\frac{V_{\text{in}} - V_C}{V_{\text{in}}}\right) e^{-AV_C}$$

$$-(V_{\text{in}})$$

$$\left(e^{\frac{-t}{(R)\cdot(C_1 - C_2 V_{\text{in}})}}\right)\cdot V_{\text{in}} = (V_{\text{in}} - V_C)e^{-AV_C}$$

$$-(V_{\text{in}})$$

$$\left(e^{\frac{-t}{(R)\cdot(C_1 - C_2 V_{\text{in}})}}\right)\cdot V_{\text{in}} = (V_{\text{in}} - V_C)e^{-A(V_{\text{in}} - V_{\text{in}} + V_C)}$$

$$-(V_{\text{in}})$$

$$\left(e^{\frac{-t}{(R)\cdot(C_1 - C_2 V_{\text{in}})}}\right)\cdot V_{\text{in}} = (V_{\text{in}} - V_C)e^{-A(V_{\text{in}} - (V_{\text{in}} - V_C))}$$

$$-(V_{\text{in}})$$

$$\left(e^{\frac{-t}{(R)\cdot(C_1 - C_2 V_{\text{in}})}}\right)\cdot V_{\text{in}} = (V_{\text{in}} - V_C)e^{-AV_{\text{in}} + A(V_{\text{in}} - V_C)}$$

$$-(V_{\text{in}})$$

$$\left(e^{\frac{-t}{(R)\cdot(C_1 - C_2 V_{\text{in}})}}\right)\cdot V_{\text{in}} = (V_{\text{in}} - V_C)e^{-AV_{\text{in}}}e^{A(V_{\text{in}} - V_C)}$$

$$-(V_{\text{in}})$$

$$\left(e^{\frac{-t}{(R)\cdot(C_1 - C_2 V_{\text{in}})}}\right)\cdot V_{\text{in}}\cdot\left(\frac{1}{e^{-AV_{\text{in}}}}\right) = (V_{\text{in}} - V_C)e^{A(V_{\text{in}} - V_C)}$$

$$-(V_{\text{in}})$$

$$\left(e^{\frac{-t}{(R)\cdot(C_1 - C_2 V_{\text{in}})}}\right)(V_{\text{in}})\left(\frac{1}{e^{-AV_{\text{in}}}}\right)(A) = (A)(V_{\text{in}} - V_C)e^{A(V_{\text{in}} - V_C)}$$

Isolating

$$W\left(\left(e^{\frac{-t}{(R)\cdot(C_1-C_2V_{\text{in}})}}\right)(V_{\text{in}})\left(\frac{1}{e^{-AV_{\text{in}}}}\right)(A)\right) = (A)(V_{\text{in}} - V_C)$$

$$-(V_{\text{in}})$$

$$W\left(\left(e^{\frac{-t}{(R)\cdot(C_1-C_2V_{\text{in}})}}\right)(V_{\text{in}})\left(\frac{1}{e^{-AV_{\text{in}}}}\right)(A)\right)\left(\frac{1}{A}\right) = V_{\text{in}} - V_C$$

$$-(V_{\text{in}})$$

$$W\left(\left(e^{\frac{-t}{(R)\cdot(C_1-C_2V_{\text{in}})}}\right)(V_{\text{in}})\left(\frac{1}{e^{-AV_{\text{in}}}}\right)(A)\right)\left(\frac{1}{A}\right) - V_{\text{in}} = -V_C$$

$$-(V_{\text{in}})$$

$$V_{\text{in}} - W\left(\left(e^{\frac{-t}{(R)\cdot(C_1-C_2V_{\text{in}})}}\right)(V_{\text{in}})\left(\frac{1}{e^{-AV_{\text{in}}}}\right)(A)\right)\left(\frac{1}{A}\right) = V_C$$

$$-(V_{\text{in}})$$

$$V_{\text{in}} - W\left(\left(e^{\frac{-t}{RC_1-RC_2V_{\text{in}}}}\right)\left(\frac{AV_{\text{in}}}{e^{-AV_{\text{in}}}}\right)\right)\cdot\left(\frac{1}{A}\right) = V_C$$

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